When and How Should the Agent Intervene? Designing an RL-Driven Multimodal Virtual Agent for Proactive Facilitation

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Abstract. Interactive AI agents and social robots designed to facilitate productive group interactions remain an emerging research area within human-AI interaction. Existing AI-mediated facilitation systems rely predominantly on rule-based, threshold-driven logic, while recent LLM-based approaches show promise in improving response fluency; however, they still lack adaptive intervention timing and multimodal grounding. This paper presents a proactive virtual AI facilitator that learns when and how to intervene during online collaborative discussions. Preliminary evaluations demonstrate the system's ability to identify and address conversational breakdowns, participation imbalances, and shallow discourse trajectories.

Keywords: Group Interaction · Proactive Facilitation · Multimodal Interaction · Large Language Model · Reinforcement Learning .

1 Introduction

Designing AI agents to support productive group interaction is an emerging research topic in both the Human-Computer Interaction (HCI) and Human-Robot Interaction (HRI) domains. In particular, collaborative learning settings require sustained engagement, balanced participation, and in-depth discussion, all of which are difficult to achieve without active facilitation [3]. Recent studies show that group interactions pose challenges beyond dyadic interaction, including tracking multiple speakers, detecting participant engagement, and generating socially appropriate behaviors. At the same time, generative AI systems in collaborative workflows remain largely individual-focused and function more as shared tools than as adaptive social participants [10, 13]. Despite growing interest in AI-mediated facilitation, existing systems primarily rely on rule-based or threshold-driven logic [12]. Although LLM-based facilitators show promise in improving response fluency, they still lack adaptive intervention timing and fail to incorporate multimodal signals beyond basic turn-taking mechanics [1].

This paper aims to addresses these gaps by introducing a proactive virtual facilitator that learns when and how to intervene in online collaborative discussions. Our contributions are: (1) a multimodal reasoning framework integrating visual–language model–based social perception, structured discourse graphs, and latent inner-thought generation to ground facilitation actions in both visual and linguistic context; and (2) a learned intervention policy, where we introduce an RL-driven proactive virtual facilitator that models group conversation as an event-driven Markov Decision Process (MDP), enabling the agent to learn intervention timing and action selection that balances engagement, participation equity, and conversational depth.

2 Related Work

2.1 AI-Mediated Group Facilitation and Intervention Strategies

Early work establishes the social value of AI agents in prompting discussion and fostering connection across structured group settings, including MOOCs, museums, and remote collaboration [14, 11, 4, 5]. However, the dominant paradigm across these systems is rule-based and heuristic: agents typically trigger predefined interventions when simple thresholds are crossed [12]. Hybrid approaches that pair LLMs with flow control [2] improve response quality but inherit this rigidity, resulting in poor intervention timing and agents that behave more like active participants than true facilitators. More recent systems utilize heuristic scoring to rank strategies or trigger proactive actions [9, 8]. To the best of our knowledge, current literature has yet to explore frameworks that jointly learn when and how to intervene using data-driven policies, leaving an opportunity to develop adaptive action selection mechanisms that respond to implicit social norms.

2.2 Multimodal Perception for Socially Aware Interaction

Integrating visual signals into conversational AI consistently improves contextual grounding and social appropriateness [7, 6]. In multi-party settings, multimodal cues are often combined for turn-taking management and speaker identification [1]. Despite these advances, visual signals in group facilitation remain narrowly applied to conversational mechanics and simple engagement classification rather than informing higher-level facilitation reasoning. While continuous latent reasoning improves proactive participation [8], it has yet to incorporate multimodal grounding. Consequently, there remains an opportunity to integrate visual engagement cues directly into the reasoning process, allowing visual perception to shape higher-level facilitation strategies rather than just conversational mechanics.

3 Methodology

We propose a proactive virtual facilitator that monitors multimodal signals in collaborative meetings to improve engagement and conversational depth. The proposed apparoch, as presented in Figure 1, operates through three tightly coupled components: multimodal perception, inner-thought reasoning, and Reinforcement Learning (RL)-based action selection.

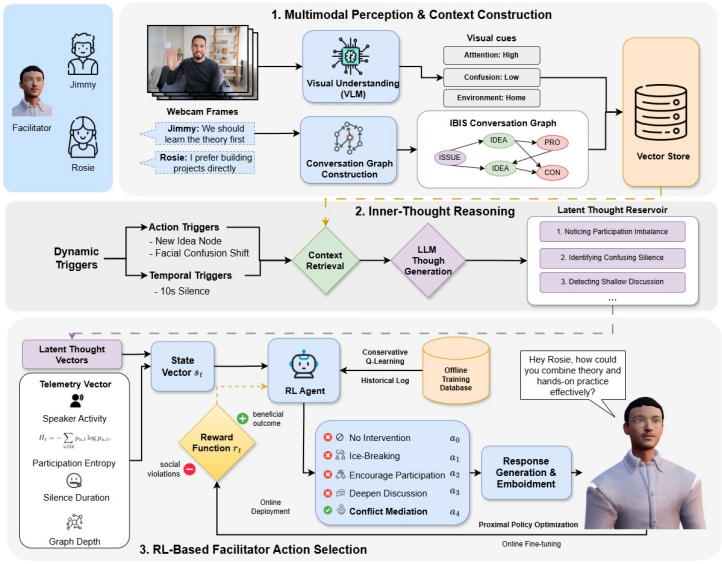


Fig. 1. Overview of the proactive virtual facilitator pipeline.

3.1 Multimodal Perception and State Construction

The system integrates heterogeneous signals into a structured representation. Participants' video frames are processed via a Vision–Language Model (GPT-4o) to extract probabilistic engagement indicators (e.g., attention, distraction, confusion). Concurrently, discourse structure is modeled using the Issue-Based Information System (IBIS) framework [5]. A fine-tuned DistilBERT classifier maps spoken transcripts into four node types (Issue, Idea, Pro, Con), linking them chronologically to form an evolving conversation graph $\mathcal{G}_t$. These visual and structural cues are aggregated into a dense meeting state $\mathcal{M}_t$ and stored in a vector database.

3.2 Inner-Thought Reasoning

An LLM-driven inner-thought module continuously reasons about when and how to intervene without mapping inputs directly to outputs. Prompted by dynamic triggers, including structural graph events, significant visual shifts, and temporal lulls, the module retrieves relevant contextual history and current visual states. Acting as a cognitive scratchpad, the LLM generates a series of covert, single-sentence thoughts assessing potential concerns like participation imbalance or shallow exploration. These thoughts are mean-pooled into a dense vector $\boldsymbol{\tau}_t$, serving as the semantic input for the downstream action-selection module.

3.3 RL-Based Facilitator Action Selection

Given the latent thought representation $\boldsymbol{\tau}_t$, an RL agent frames the conversation as an event-driven Markov Decision Process to select from five discrete actions: a_0 (no intervention), or a_{1-4} (ice-breaking, encouraging participation, deepening discussion, and conflict mediation). The state space $s_t =$

$[\boldsymbol{\tau}_t \parallel \mathbf{z}_t]$ concatenates the semantic thought vector with a telemetry vector $\mathbf{z}_t = [\Delta_t^{\text{sil}}, H_t, b_t, d_t]$. This captures silence duration (Δ_t^{sil}), speaker activity (b_t), graph depth (d_t), and participation entropy H_t:

$$H_t = -\sum_{u \in \mathcal{U}} p_{u,t} \log p_{u,t}, \tag{1}$$

where $p_{u,t}$ is the proportion of speaking time for participant u.

The reward function r_t evaluates objective proxies over a 30-second window (Δt), penalizing interruptions while rewarding measurable improvements like elicited responses or positive visual engagement shifts. To ensure safe deployment, the policy is pre-trained offline via Conservative Q-Learning (CQL) on historical logs. Online, the selected action prompts an LLM to generate the final natural-language response, and the policy is periodically fine-tuned via Proximal Policy Optimization (PPO) using newly collected transition tuples to adapt to group dynamics.

4 Conversation Platform and Interaction Scenarios

4.1 Online Conversation Platform with a Virtual Facilitator.

We developed a custom web-based video conferencing platform utilizing React and WebRTC. The interface features a 3D virtual facilitator rendered in-browser via WebGL, surrounded by participant video feeds. The platform operates on a real-time loop: raw multimodal signals, including continuous voice audio and structural meeting events, are streamed to a FastAPI backend via WebSocket connections. Once the inner-thought and RL modules determine an intervention, the generated text-to-speech audio and corresponding lip-sync cues are transmitted back to the client. The 3D avatar then embodies this response dynamically using procedural morph targets for synchronized facial expressions and speech animation, seamlessly integrating the AI pipeline into the live meeting environment.

4.2 Demonstration Scenarios.

To demonstrate the system's facilitation capabilities, Figure 2 illustrates two interaction scenarios observed during testing. In each case, the virtual facilitator monitors multimodal signals and selectively intervenes during a collaborative discussion.

Scenario 1: Restarting a Stalled Discussion. Participants discuss the ethical implications of AI in hiring. Initially, the participation entropy H_t of the two users remains stable as both contribute equally, and the conversation graph $\mathcal{G}_t$ continues to expand with new nodes, indicating a healthy and productive exchange; thus, the agent selects a_0 (no intervention). Later, as the topic reaches a shallow point, the users begin to fall silent more frequently. This extended silence triggers the temporal threshold and also the inner-thought module infer from the stagnant discourse graph that the conversation lacks depth and is nearing a premature end. Recognizing this loss of conversational momentum, the agent executes a_1 (ice-breaking). The facilitator introduces a contextual prompt grounded in the prior discussion, asking about personal experiences with automated misjudgment, which successfully re-engages the users.

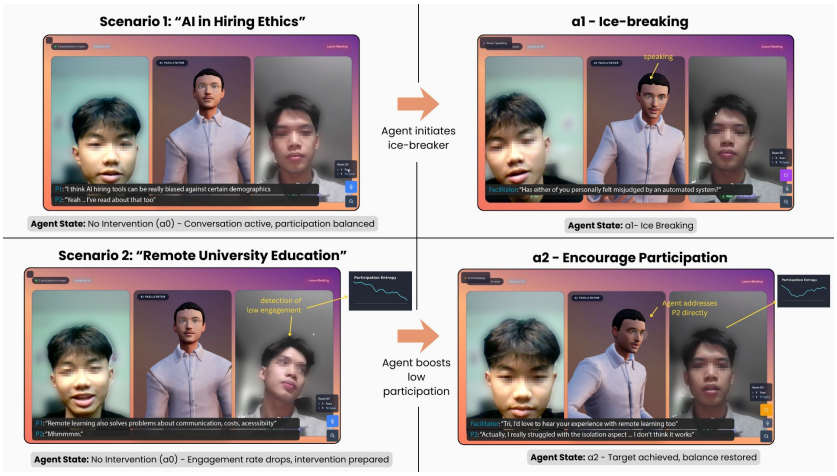


Fig. 2. Examples of adaptive intervention by the virtual facilitator.

Scenario 2: Rebalancing an Uneven Exchange. During a discussion on remote university education, participation entropy of the second user decreases steadily as first participant dominates the speaking time. The agent initially maintains a_0, but as the imbalance grows, the VLM detects low engagement cues such as gaze aversion from the quieter participant. Identifying this imbalance, the agent selects a_2 (encourage participation) and addresses the quieter user directly ("Tri, I would love to hear your experience"). This targeted intervention elicits a new substantive contribution, restores conversational balance, and introduces a new perspective to the discourse graph.

5 Ongoing Work

5.1 Evaluation Across Different Facilitation Conditions

Our current work focuses on validating the proposed approach across different facilitation conditions, including a no-facilitator baseline and a virtual agent condition, as presented in Fig. 2. We will also investigate the impact of physical embodiment through social robots, such as Furhat and Reachy Mini, equipped with facial expressions and gaze behaviors to support verbal communication. The study will combines quantitative metrics, including participation entropy, graph depth, and engagement scores, with subjective evaluations capturing perceived naturalness, trust, and facilitation effectiveness. This setup enables us to directly examine how physical embodiment interacts with the underlying facilitation policy, isolating the added value of robotic presence beyond algorithmic intervention.

5.2 Generalization Across Interaction Scenarios

In parallel, we are evaluating the facilitator's behavior across a wide range of real-world interaction scenarios, including collaborative task-oriented discussions, social conversations, and cross-cultural group interactions. These settings expose the agent to varying conversational norms, such as differences in turn-taking, silence tolerance, and communication style. We operationalize scenario-

specific objectives through targeted adjustments to the reward function and action space, allowing the same RL policy to be stress-tested under shifting interaction dynamics. This ongoing analysis examines the extent to which a policy trained in one context maintains effectiveness in others, and where adaptation or fine-tuning becomes necessary.

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